

Math

MATHEMATICS (GENERAL EQUATION)

FRACTION

Definition: A fraction is defined as ratio or quotient of 2 numbers.

Example: $\frac{3}{5}$, $\frac{4}{11}$, $\frac{10}{3}$, $\frac{18}{18}$

In a fraction $\frac{a}{b}$, (a = numerator b = denominator). If a is less than b ($a < b$), $\frac{a}{b}$ is proper fraction.

Example: $\frac{5}{8}$, $\frac{7}{11}$, $\frac{30}{101}$

If a is greater than or equal to b ($a > b$), $\frac{a}{b}$ is an improper fraction.

Example: $\frac{8}{5}$, $\frac{100}{5}$, $\frac{11}{11}$

Definition: A mixed number is defined as the sum of whole number and a fraction.

Example: $5\frac{3}{4} = 5 + \frac{3}{4}$ $11\frac{5}{7} = 11 + \frac{5}{7}$

Convert the following to mixed numbers (use Law of Division).

Example: 1. $\frac{8}{5} = 1\frac{3}{5}$ 2. $\frac{11}{2} = 5\frac{1}{2}$ 3. $\frac{18}{7} = 2\frac{4}{7}$

Convert the following to improper fractions.

Example: 1. $4\frac{2}{3} = \frac{14}{3}$ 2. $8\frac{1}{4} = \frac{33}{4}$ 3. $10\frac{3}{5} = \frac{53}{5}$

Definition: The Least Common Multiple (LCM) of 2 numbers (a and b) is the smallest number that is divisible by a and divisible by b at the same time.

Example: Find the LCM of the following.

1. 18 and 24

Method 1: Prime Factorization

$$18 = 2 \cdot 3^2$$

$$24 = 2^3 \cdot 3$$

The LCM, therefore, will be the product of the greatest powers of each kind of factors.

$$\text{LCM} = 2^3 \cdot 3^2 = 8 \cdot 9 = 72$$

**Method 2: Division Method**

$$2 / 18, 24$$

$$2 / 9, 12$$

$$2 / 9, 6$$

$$3 / 9, 3$$

$$3 / 3, 1$$

$$/ 1, 1$$

$$\text{LCM} = 2^3 \cdot 3^2 = 8 \cdot 9 = 72$$

2.) 15 and 36

Method 1: Prime Factorization

$$15 = 3 \cdot 5$$

$$36 = 2^2 \cdot 3^2$$

$$\text{LCM} = 2^2 \cdot 3^2 \cdot 5 = 180$$

Method 2: Division Method

$$2 / 15, 36$$

$$2 / 15, 18$$

$$3 / 15, 9$$

$$3 / 5, 3$$

$$5 / 5, 1$$

$$/ 1, 1$$

$$\text{LCM} = 2^2 \cdot 3^2 \cdot 5 = 180$$

3.) 8 and 18

$$\text{LCM} = 18$$

$$\text{LCM} = 11(8) = 88$$

4. 11 and 8

$$\text{LCM} = 11(8) = 88$$

The **LCM** of 2 numbers that are relatively prime to each other (**Example:** 11 and 8) is the product of the given 2 numbers.

Addition of Fraction

Examples:

$$1. \quad \frac{3}{4} + \frac{1}{4} = \frac{3+1}{4} = \frac{4}{4} = 1$$

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$$2. \frac{5}{11} + \frac{7}{11} + \frac{4}{11} = \frac{5+7+4}{11} = \frac{16}{11} = 1\frac{5}{11}$$

$$3. 2\frac{3}{5} + 4\frac{1}{5} = (2+4) + \left(\frac{3}{5} + \frac{1}{5}\right) = 6 + \frac{4}{5} = 6\frac{4}{5}$$

$$4. \frac{5}{8} + \frac{7}{12} + \frac{3}{4} = \frac{3(5) + 2(7) + 6(3)}{24} = \frac{47}{24} = 1\frac{23}{24}$$

$$5. 4\frac{3}{4} + 5\frac{2}{5} = (4+5) + \left(\frac{3}{4} + \frac{2}{5}\right) = 9 + \frac{5(3) + 4(2)}{20} = 9 + \frac{23}{20} = 9 + 1\frac{3}{20} = 10\frac{3}{20}$$

Subtraction of Fraction

Examples:

$$1. \frac{5}{7} - \frac{2}{7} = \frac{5-2}{7} = \frac{3}{7}$$

$$2. 8\frac{6}{11} - 4\frac{3}{11} = (8-4) + \left(\frac{6}{11} - \frac{3}{11}\right) = 4 + \frac{3}{11} = 4\frac{3}{11}$$

$$3. \frac{5}{8} - \frac{3}{5} = \frac{5(5) - 8(3)}{40} = \frac{1}{40}$$

$$4. 8 - \frac{2}{7} = \frac{7(8) - 1(2)}{7} = \frac{54}{7} = 7\frac{5}{7}$$

$$5. 6\frac{1}{3} - 3\frac{1}{8} = (6-3) + \left(\frac{1}{3} - \frac{1}{8}\right) = 3 + \frac{8(1) - 3(1)}{24} = 3 + \frac{5}{24} = 3\frac{5}{24}$$

Definition: The *Greatest Common Factor/Divisor* (**GCF** or **GCD**) of two numbers (*a* and *b*) is the largest factor or divisor common to both *a* and *b*.

Example: Find the **GCF** or **GCD** of the following.

1. 18 and 24

Method 1: List all the divisors of 18 and 24.

Divisors or Factors of 18 = {1, 2, 3, 6, 9, 18}

Divisors or Factors of 24 = {1, 2, 3, 4, 6, 8, 12, 24}

Common Divisors = {1, 2, 3, 6}

GCD/GCF = 6

**Method 2: Prime Factorization**

$$18 = 2 \cdot 3^2$$

$$24 = 2^3 \cdot 3$$

The **GCD/GCF**, therefore, will be the product of the least powers of each kind of factors.

$$\text{GCD/GCF} = 2 \cdot 3 = 6$$

2.) 15 and 36

Method 1: List all the divisors of 15 and 36.

Divisors or Factors of 15 = {1, 3, 5}

Divisors or Factors of 36 = {1, 2, 3, 4, 6, 9, 12, 18, 36}

Common Divisors = {1, 3, 5}

$$\text{GCD/GCF} = 3$$

Method 2: Prime Factorization

$$15 = 3 \cdot 5$$

$$36 = 2^2 \cdot 3^2$$

$$\text{GCD/GCF} = 3$$

3.) 8 and 8

$$\text{GCD/GCF} = 8$$

2.) 12 and 13

$$\text{GCD/GCF} = 1$$

Note: The **GCD/GCF** of 2 numbers which are relatively prime to each other is 1, like 12 and 13.

Multiplication of Fractions

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

Example:

$$1. \quad \frac{21}{8} \cdot \frac{16}{3} = \frac{336}{24} = 14 \quad \text{Or} \quad \frac{21}{8} \cdot \frac{16}{3} = \frac{21}{3} \cdot \frac{16}{8} = 7(2) = 14$$

$$2. \quad \frac{18}{25} \cdot \frac{5}{24} = \frac{90}{600} = \frac{90 \div 30}{600 \div 30} = \frac{3}{20} \quad (30 \text{ being the GCD of } 90 \text{ and } 600)$$

Or by cancellation process using the principle of **GCD**

$$\frac{18}{25} \cdot \frac{5}{24} = \frac{3}{5} \cdot \frac{1}{4} = \frac{3}{20}$$



Division of Fractions

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$$

Example:

$$1. \quad \frac{16}{27} \div \frac{8}{36} = \frac{16}{27} \cdot \frac{36}{8} = \frac{2}{3} \cdot \frac{4}{1} = \frac{8}{3} = 2\frac{2}{3}$$

$$2. \quad 1\frac{3}{4} \div 2 = \frac{7}{4} \cdot \frac{1}{2} = \frac{7}{8}$$

$$3. \quad 1\frac{7}{8} \div 8\frac{1}{4} = \frac{15}{8} \div \frac{33}{4} = \frac{15}{8} \cdot \frac{4}{33} = \frac{5}{2} \cdot \frac{1}{11} = \frac{5}{22}$$

$$4. \quad \left(\frac{8}{9} - \frac{1}{3}\right) \div \left(\frac{3}{4} - \frac{1}{6}\right) = \left(\frac{8-3}{9}\right) \div \left(\frac{9-2}{12}\right) = \frac{5}{9} \cdot \frac{12}{7} = \frac{5}{3} \cdot \frac{4}{7} = \frac{20}{21}$$

Applications:

Examples:

1. In budgeting their income, a young couple allows $\frac{1}{6}$ for rent, $\frac{1}{2}$ for food and the rest for clothing. What fractional part do they provide for clothing?

$$\text{Solution: Clothing} = 1 - \left(\frac{1}{6} + \frac{1}{2}\right) = 1 - \frac{2}{3} = \frac{1}{3}$$

2. A landowner who owns $25\frac{1}{2}$ hectares of land sold $8\frac{3}{4}$, $4\frac{1}{3}$ and 6 hectares of land during the months of January, February and March, respectively. How many hectares of land does he own now?

Solution: Number of hectares the landowner owns now

$$= 25\frac{1}{2} - \left(8\frac{3}{4} + 4\frac{1}{3} + 6\right)$$

$$= 25\frac{1}{2} - 19\frac{1}{12} = 6\frac{5}{12}$$



3. How many pieces of $7\frac{1}{2}$ inch wire can be cut from $31\frac{1}{4}$ ft. roll of wire?

Solution: $31\frac{1}{4} \text{ ft.} = \frac{125}{4} \times 12 \text{ inches} = 375 \text{ inches}$
 $375 \text{ inches} \div 7.5 \text{ inches} = 50 \text{ pieces.}$

4. A painter estimates that it will require 75 hrs to paint a certain house. If he works 7 hrs. a day, how many days will it take him to paint the house?

Solution: $75 \div 7.5 = 10 \text{ days}$

5. If a coal cost P3, 550.50 per ton, find the cost of $4\frac{3}{4}$ tons.

Solution: $P3,550.50 \times 4.75 = P16,864.875$

6. A woman bought 9.5 yd. of dress material at P305.00 per yd. and $4\frac{1}{2}$ yd. of ribbons at P37.50 per yd. What was the total amount that she paid?

Solution: $9.5 \times P305.00 = P2,897.50$
 $4.5 \times P37.50 = P168.75$
Total amount paid = P3,066.25

RATIO AND PROPORTION

Definition: A *ratio* is a certain relation between 2 numbers expressed in terms of a quotient.

Example:

1. $3:5 =$ "3 is to 5"

$$3:5 = 3 \div 5 = \frac{3}{5}$$

2. A certain organization has 36 men and 15 women. Find the men-women ratio.

Solution: $36:15 = \frac{36}{15} = \frac{12}{5} = 12:5$

3. A rod, 72 inches long is to be cut into 2 pieces in the ratio 4:5. What is the length of the shorter piece?

Solution: Shorter piece = $\frac{4}{9}(72) = 32 \text{ inches}$

Definition: A *proportion* is an equality between 2 ratios.

$a:b = c:d$ is a proportion if and only if $ad = bc$. That is the product of the extremes (a and d) is equal to the product of the means (b and c).

$2:3 = 6:9$ is a proportion because,

$$2(9) = 3(6) = 18$$

$3:4 = 9:8$ is not proportion because,

$$3(8) \neq 4(9), \text{ that is } 24 \neq 36.$$

$$a:b = c:d \text{ is equivalent to } \frac{a}{b} = \frac{c}{d}$$

$$\begin{array}{c} bc \\ a:b = c:d \\ \boxed{\times} \\ ad \end{array}$$

Find the missing term in each of the following.

1. $42:x = 6:7$

$$6x = 42(7)$$

$$6x = 294$$

$$x = 49$$

$$\begin{array}{r} 6x = 294 \\ \underline{6} \quad \underline{6} \\ x = 49 \end{array}$$

2. $\frac{5}{4} = \frac{10}{x}$

$$5x = 40$$

$$x = 8$$

3. $x:\frac{1}{2} = 2:\frac{3}{4}$

$$\frac{3}{4}x = \frac{1}{2}(2)$$

$$\frac{3}{4}x = 1$$

$$x = \frac{4}{3}$$

4. $\frac{4}{9} = 2x$

$$18x = 4$$

$$x = \frac{4}{18} = \frac{2}{9}$$

$$\frac{4}{9} = \frac{2 \cdot 2}{1 \cdot 9}$$

$$4:9 = 4:9$$

Applications

Examples:

1. An investment of P15,000 results in a gain of P3,000. How much must be invested to earn P5,000?

Solution: $\frac{P15,000}{P3,000} = \frac{x}{P5,000}$

$$x = P25,000$$

2. A motorcycle covers 225 km. in 2.5 hrs. How far can he do in 1.75 hrs?

Solution: $\frac{225}{2.5} = \frac{x}{1.75}$

$$x = 157.5 \text{ km}$$



2. The interior angles of a triangle are in the ratio of 2:3:4. What are the measurements of the 3 angles?

Solution: The sum of the interior angles of any triangle is equal to 180° .

$$2x + 3x + 4x = 180^\circ$$

$$9x = 180^\circ$$

$$x = 20^\circ$$

$$2x = 40^\circ$$

$$3x = 60^\circ$$

$$4x = 80^\circ$$

Therefore, the measurements of the 3 angles are 40° , 60° and 80° .

PERCENTAGE

Definition: *Percent* is defined as an expression which indicates the number of parts taken from a hundred. *Per* is a Latin word meaning "for each" or "for every", while *cent* means a hundred. The symbol used to express percent is %.

Conversion Techniques

A. Changing Percent to Decimal

To change a percent to a decimal, simply move the decimal point two places to the left and drop the percent sign.

Example:

1. $55\% = 0.55$

2. $0.08\% = 0.0008$

3. $\frac{1}{4}\% = 0.25\% = 0.0025$

B. Changing Decimal to Percent

To change decimal to percent, move the decimal point two places to the right and add the percent sign.

Example:

1. $0.38 = 38\%$

2. $0.4123 = 41.23\%$

3. $0.0062 = 0.62\%$

C. Changing Percent to Fraction

To change a percent to fraction, drop the percent sign and replace by 100 as denominator. If the percent is in decimal, move the decimal point two places to the left after dropping the percent sign and convert the decimal to its fractional equivalent.

Example:

$$1. \quad 36\% = \frac{36}{100} = \frac{9}{25}$$

$$2. \quad 7.5\% = 0.075 = \frac{75}{1000} = \frac{3}{40}$$

$$3. \quad 0.84\% = 0.0084 = \frac{84}{10,000} = \frac{21}{2,500}$$

D. Changing Fraction to Percent

To change a fraction to a percent, perform the indicated division, and then move the decimal point of the quotient two places to the right and add the percent sign.

Example:

$$1. \quad \frac{3}{4} = 0.75 = 75\%$$

$$2. \quad 3\frac{1}{2} = 3.5 = 350\%$$

$$3. \quad \frac{1}{8} = 0.125 = 12.5\%$$

More Problems

1. 40% of P200 is how much?

$$\begin{aligned} \text{Solution: } 40\% [P200] &= N \\ 0.4 (P200) &= N \\ P80 &= N \end{aligned}$$

2. What percent of P150 is P7.50?

$$\text{Solution: } R (P150) = P7.50$$

$$R = \frac{P7.50}{P150} = 0.05 = 5\%$$



3. What part of 30 is 180?

Solution: $R(30) = 180$
 $R = 6 = 600\%$

4. 15% of P150 is what percent of 30% of P3,000?

Solution: $15\% \text{ of } P150 = 0.15 (P150) = P22.50$
 $30\% \text{ of } P3,000 = 0.30 (3,000) = P900$
P22.50 is what percent of P900
 $P22.50 = R (P900)$

$$\frac{P22.50}{P900} = R = 0.025 = 2.5\%$$

5. P200 increased by 10% is how much?

Solution: $N = P200 + 0.10 (P200) = P200 + P20$
 $N = P220$

6. The gross profit of Ana's sari-sari store last year was 40% of the total sales. How much was the gross profit if total sales were P480,500?

Solution: $\text{Gross profit} = 40\% (P480,000) = P192,200$

7. Ann, cashier of ABC Department store is given 15% discount on cash purchase. How much will her discount be if she buys 2 dresses at P180 each and one pair of shoes costing P650?

Solution: $P180 \times 2 \text{ dresses} = P360$
Pair of shoes = $\frac{650}{}$
Total purchase = $\frac{P1,010}{}$
Total discount = $0.15 (P1,010) = P151.50$

8. Student population in a certain Barangay High School is composed of 2,300 females and 1,900 males.

A. What percent of the population are males?

B. What percent are females?

Solution: $\text{Total population} = 2,300 + 1,900 = 4,200$

$$\text{Male \%} = \frac{1,900}{4,200} = 45.24\%$$

$$\text{Female \%} = \frac{2,300}{4,200} = 54.76\%$$

DIVISIBILITY OF NUMBERS

Rules of Divisibility of Numbers.

1. A number is divisible by 2 if it ends in 0, 2, 4, 6, or 8. An even number is divisible by 2.

Example: 324

50, 630

4, 356

2. A number is divisible by 3 if the sum of the digits is divisible by 3.

Example: 459 is divisible by 3 because the sum of its digits is 18 and 18 is divisible by 3. 13, 462 and 2, 352 are also divisible by 3.

3. A number is divisible by 4 if the number formed by the last 2 digits is divisible by 4.

Example: 43, 236 is divisible by 4 because the number formed by the last two digits is 36 and 36 is divisible by 4. 48, 324 and 53, 000 are also divisible by 4.

4. A number is divisible by 5 if it ends in 0 or 5.

Example: 5, 035

23, 210

5. A number is divisible by 6 if it is divisible by 2 and divisible by 3.

Example: 4, 452

32, 478

6. Divisibility by 7

Rule. Double the last digit and subtract it from the rest of the numbers, if the result is divisible by 7, then the original number is divisible by 7.

Example: 252 Double the last digit ($2 + 2 = 4$).
Subtract 4 from 25, the result is 21 and since 21 is divisible by 7, therefore 252 is divisible by 7.
2, 282 is also divisible by 7.

7. A number is divisible by 8 if the number formed by the last 3 digits is divisible by 8.

Example: 11, 648 since 648 is divisible by 8,
11, 648 is therefore divisible by 8.

8. A number is divisible by 9 if the sum of the digits is divisible by 9.

Example: 45, 927 is divisible by 9 because the sum of its digits is 27 and 27 is divisible by 9.

Exercises: Find the value of k which satisfies the given condition.

1. The number $432k$ is divisible by 5 if k is,
A. 2 B. 3 C. 4 D. 0
Answer: **d. 0**
2. The number $532k6$ is divisible by 6 if k is,
A. 4 B. 1 C. 2 D. 0
Answer: **c. 2**
3. The number $7456k8$ is divisible by 4 if k is,
A. 5 B. 4 C. 3 D. 1
Answer: **b. 4**
4. The number $k96$ is divisible by 7 if k is,
A. 2 B. 1 C. 4 D. 5
Answer: **b. 1**
5. The number $48k$ is divisible by 8 if k is,
A. 1 B. 5 C. 0 D. 3
Answer: **c. 0**

NUMBER SERIES

Each of the items contains a series of numbers which appear according to a certain rule or order. Relate the series from suggested answer and determine which number is next in the series.

1. - 4, - 7, - 10, - 13, Answer: - 16
2. 3, 6, 12, 24, 48, Answer: 96
3. $\frac{3}{2}, \frac{1}{2}, \frac{1}{6}, \text{---}, \frac{1}{54}$ Answer: $+\frac{1}{18}$
4. 95, 88, 79, 68, —, 40 Answer: 55
5. A, E, I, M, Q, Answer: U

Operations on Real Numbers

A. Addition of Real Numbers

1. To add numbers with the same sign, first add their absolute values. The sign of the answer (either + or -) is the same as the sign of the two numbers.

Example: $-8 + (-4) = -(8 + 4) = -12$

$(+12) + (+6) = 12 + 6 = 18$

-12 and 18 are called sums.

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2. To add two numbers with different signs, first subtract their absolute values. The answer is positive if the positive number has the larger absolute value. The answer is negative if the negative number has the larger absolute value.

Example: $-8 + (+4) = -4$

$8 + (-4) = 4$

B. Subtraction of Real Numbers

Rule: For all real numbers a and b , $a - b = a + (-b)$. That is, change the sign of the second number then add. The result of subtraction is called difference.

Example: $10 - 8 = 10 + (-8) = 2$

$-12 - (-7) = -12 + (+7) = -5$

C. Multiplication of Real Numbers

Rule: The product of 2 numbers with same sign is positive. The product of 2 numbers with different signs is negative.

Example: $(-8)(-5) = 40$

$(-8)(5) = -40$

D. Division of Real Numbers

Rule: The quotient of 2 numbers with same sign is positive. The quotient of 2 numbers with different signs is negative.

Example: $30 \div -6 = \frac{30}{-6} = -5$

$-24 \div -4 = \frac{-24}{-4} = 6$

Note: For all real numbers a and b where $b \neq 0$, $\frac{a}{b} = a\left(\frac{1}{b}\right)$. There is no

reciprocal for the number 0, so division by 0 is not defined. Example $\frac{5}{0}$ is not defined.

Polynomials

Important Terms

1. Algebraic expression is any collection of variables, constants, grouping symbols and symbols of operation.

Example:

$$3x^2y, \quad -2x + y, \quad \pi y^2$$

$$\sqrt{x + 2y}, \quad \frac{x + y}{x - y}$$

2. Term is any constant, variable or indicated product, quotient or roots of constants and variables. Terms in algebraic expression are separated by plus or minus.

Example:

a) In $4x^2 + 3x + 5$, there are 3 terms namely $4x^2$, $3x$ and 5 .

b) In $3y^2 - \frac{2t}{z}$, there are 2 terms namely $3y^2$ and $-\frac{2t}{z}$.

3. A coefficient is a factor in a term.

Example:

In the algebraic expression, $5x^3 - 4x^2y^2 + 3y^3$, $-4x^2y^2$ is a term

where -4 is the numerical coefficient of x^2y^2 , x^2 is the coefficient of $-4y^2$

and y^2 is the coefficient of $-4x^2$.

4. Polynomial is an algebraic expression consisting of one or more terms that have real number coefficients, all variables are raised to natural number exponents, and the operations involving variables are limited to addition, subtraction and multiplication.

Examples of Polynomials:

a. $5 - 3x$

b. $6x^2y^3 - 3xy^2 + 3$

c. $-2xy$

d. $\sqrt{2}xy - \frac{1}{2}x + 3$

Examples of not polynomials:

a. $\frac{2}{x} - \frac{3}{y}$

b. $x^{-3} - 2x^{-2} - 3x^{-1} + 3$

5. Degree of a term with one variable is the exponent of the variable.

Example:

a. 3 is the degree of $-2t^3$

b. 6 is the degree of $5y^6$

c. 0 is the degree of 3, (since 3 can be written as $3x^0$)

Note: degree of a constant is zero.

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Degree of a term with 2 or more variables is the sum of the exponents of the variables.

Example:

- a. 5 is the degree of $3x^2y^3$
- b. 7 is the degree of $-2x^2y^2z^4$

6. Degree of a polynomial is the highest degree of any of the terms in the polynomial.

Example:

- a. 3 is the degree of $5a^3 - 3a^2 + 2$
- b. $3x^3 - 2x + \frac{1}{x}$ has no degree, since the given algebraic expression is not a polynomial.

7. Like terms or similar terms have the same variables to the same power.

Example:

- a. $-2x$, $5x$ and $\frac{1}{2}x$ and are like terms
- b. $3xy^2$ and $-5xy^2$ and are like terms
- c. $2x^2y$ and $5xy^2$ and are not like terms

Addition of Polynomials

Rule: Add similar terms together.

Example:

1. Add $5x^2 + 3xy - 2y^2$, $3x^2 - 2xy + 5y^2$ and $-4x^2 + 2xy - 3y^2$.

Solution:

$$\begin{array}{r}
 5x^2 + 3xy - 2y^2 \\
 3x^2 - 2xy + 5y^2 \\
 -4x^2 + 2xy - 3y^2 \\
 \hline
 4x^2 + 3xy \text{ (sum)}
 \end{array}$$

2. Add $-3a^2b + 2a^2b^2 - 5b^2$, $4a^2b^2 - 3ab^2 + 5b^2$ and $5ab^2 - 3a^2b^2 + 4b^2$.

Solution:

$$\begin{array}{r}
 -3a^2b \quad \quad + 2a^2b^2 - 5b^2 \\
 \quad - 3ab^2 + 4a^2b^2 + 5b^2 \\
 \quad \quad 5ab^2 - 3a^2b^2 + 4b^2 \\
 \hline
 -3a^2b + 2ab^2 + 3a^2b^2 + 4b^2 \text{ (sum)}
 \end{array}$$

Subtraction of Polynomials

Rule: Change the sign of the subtrahend and proceed as in addition.

Example:

Subtract $5x^2 - 3xy + 4y^2$ from $-10x^2 + 2xy - y^2$

$-10x^2 + 2xy - y^2$ (minuend)

$5x^2 - 3xy + 4y^2$ (subtrahend)

(-) (+) (-)

$-15x^2 + 5xy - 5y^2$ (difference)

Another method $(-10x^2 + 2xy - y^2) - (5x^2 - 3xy + 4y^2)$

$$= -10x^2 + 2xy - y^2 - 5x^2 + 3xy - 4y^2$$

$$= -15x^2 + 5xy - 5y^2$$

Removal of Grouping Symbols

Rules: To remove a grouping symbol preceded by a

- Minus sign, change the sign of each of the terms.
- Plus sign, no change of sign to be made.
- Factor, use the distribution law.

Example: Simplify

$$1. -(x - y + 2) + (4x - y) - 2(x + 3y - 2)$$

$$= -x + y - 2 + 4x - y - 2x - 6y + 4$$

$$= x - 6y + 2$$

$$2. - \{ -2x - y - (x + y - 2) - 4 \}$$

$$= - \{ -2x - y - x - y + 2 - 4 \}$$

$$= - \{ -3x - 2y - 2 \}$$

$$= 3x + 2y + 2$$

Laws on Exponents

$$I. a^m \cdot a^n = a^{m+n}$$

Example: 1. $x^4 \cdot x^3 = x^{4+3} = x^7$

2. $(2a^3b)(-3a^2b^3) = -6a^5b^4$

$$II. (a^m)^n = a^{mn}$$

Example: 1. $(x^4)^3 = x^{4(3)} = x^{12}$

2. $(a^5)^2 (a^3)^4 = a^{10} (a^{12}) = a^{10+12} = a^{22}$



$$III. (abc)^n = a^n b^n c^n$$

Example: 1. $(2x)^4 = 2^4 x^4 = 16x^4$
 2. $-2^2(-x^2)^3 = -4(-x^6) = 4x^6$

$$IV. \frac{a^m}{a^n} = a^{m-n}, \text{ if } m > n$$

Example: $\frac{x^7}{x^3} = x^{7-3} = x^4$

$$\frac{a^m}{a^n} = \frac{1}{a^{n-m}}, \text{ if } n > m$$

Example: $\frac{x^3}{x^7} = \frac{1}{x^{7-3}} = \frac{1}{x^4}$

$$\frac{a^m}{a^n} = 1, \text{ if } n = m$$

Example: $\frac{x^3}{x^3} = 1$

$$V. \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

Example: $\left(\frac{2x}{3y^3}\right)^5 = \frac{2^5 x^5}{3^5 y^{15}} = \frac{32x^5}{243y^{15}}$

$$VI. a^0 = 1, a \neq 0$$

Example: 1. $5^0 = 1$
 2. $-2^0 = -1$

$$VII. a^{-n} = \frac{1}{a^n}$$

Example: 1. $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$

2. $-1^{-1} \cdot 2^{-1} = -1 \cdot \frac{1}{2} = -\frac{1}{2}$

3. $\frac{2^{-1} \cdot 3^{-1}}{2^{-1} - 3^{-1}} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2} - \frac{1}{3}} = \frac{\frac{1}{6}}{\frac{1}{6}} = 1$

$$VIII. \left(\frac{a}{b}\right)^{-n} = \frac{a^n}{b^n} = \frac{b^n}{a^n}$$

Example: 1. $\left(\frac{3}{4}\right)^{-2} = \left(\frac{4}{3}\right)^2 = \frac{16}{9}$

2. $\left(\frac{2a}{b}\right)^{-3} = \left(\frac{b}{2a}\right)^3 = \frac{b^3}{8a^3}$

3. $(x^{-2} - y^2)^{-1} = \left(\frac{1}{x^2} - \frac{1}{y^2}\right)^{-1} = \frac{x^2 y^2}{y^2 - x^2}$

MORE EXERCISES. Simplify.

1. $\frac{x+x}{x \cdot x}$ Ans. $\frac{2}{x}$

2. $\frac{-9^{4k}}{81^{2k}}$ Ans. -1

3. $\left(\frac{4x^2}{32x}\right)^3 \left(\frac{4}{x^3}\right)^3$ Ans. 1/8

4. $\frac{-x^2}{y^2} - \left(\frac{x^2 y}{y^2 x}\right)^2$ Ans. $-\frac{2x^2}{y^2}$

5. $\left(\frac{4x^2}{(6(y^3)^2 x)}\right)^2 \left(\frac{12y^6}{8x}\right)^2$ Ans. 1

6. $\frac{4^{-1} + 2^{-1}}{4^{-1} - 2^{-1}}$ Ans. -3

7. $-2^{-2} - 2^0 - 2^2$ Ans. -7/4

Math

8. $\frac{x^{-1} + y^{-1}}{x^{-2} - y^{-2}}$

Ans. $\frac{-xy}{y-x}$

9. $-(x-1) - (2-(x+1))$

Ans. 0

10. $(2x-3y)(4x+5y)$

Ans. $8x^2 - 2xy - 15y^2$

Multiplication of Polynomials

Rule: To multiply two polynomials, use distributive law and laws of exponents.

Example: Find the product

1. $(3x-y)(5x+3y) = 3x(5x+3y) - y(5x+3y)$

$$= 15x^2 + 9xy - 5xy - 3y^2$$

$$= 15x^2 + 4xy - 3y^2$$

2. $(x^2 - 2x - 4)(3x^2 + x - 2)$

$$= x^2(3x^2 + x - 2) - 2x(3x^2 + x - 2) - 4(3x^2 + x - 2)$$

$$= 3x^4 + x^3 - 2x^2 - 6x^3 - 2x^2 + 4x - 12x^2 - 4x + 8$$

$$= 3x^4 - 5x^3 - 16x^2 + 8$$

Division of Polynomials

Rule 1: To divide monomials, use laws of exponents in division.

Example: $\frac{6x^3y}{2xy^4} = \frac{3x^2}{y^3}$

Rule 2: To divide polynomial by a monomial, use $\frac{a+b}{c} = \frac{a}{c} + \frac{b}{c}$, $c \neq 0$

Example: $\frac{8a^3b - 4ab^2}{2ab} = \frac{8a^3b}{2ab} - \frac{4ab^2}{2ab} = 4a^2 - 2ab$

Rule 3: The last rule is to divide a polynomial by a polynomial with at least 2 terms. This type of division is applied only when the degree of the polynomial in the numerator is greater than or equal to the degree of the polynomial in the denominator.

1. Arrange the terms of both the dividend and the divisor in descending powers of the variable.
2. Divide the first term of the dividend by the first term of the divisor, giving the first term of the quotient.
3. Multiply each term of the divisor by the first term of the quotient and subtract the product from the dividend.
4. Use the remainder obtained in step 3 as a new dividend, and repeat Steps 2 and 3.
5. Continue the process until the remainder is reached whose degree should be less than the degree of the divisor.



For exact division with remainder 0

$$\frac{\text{dividend}}{\text{divisor}} = \text{quotient}$$

For remainder $\neq 0$

$$\frac{\text{dividend}}{\text{divisor}} = \text{quotient} + \frac{\text{remainder}}{\text{divisor}}$$

Example 1: $(x^3 + 5x^2 + 5x - 2) \div (x + 2)$

$$\begin{array}{r} x^2 + 3x - 1 \\ (x + 2) \overline{) x^3 + 5x^2 + 5x - 2} \\ \underline{-(x^3 + 2x^2)} \\ 3x^2 + 5x \\ \underline{-(3x^2 + 6x)} \\ -x - 2 \\ \underline{-(-x - 2)} \\ 0 = \text{remainder} \end{array}$$

Thus, the quotient is $x^2 + 3x - 1$

Example 2. $(5x^3 - 2x^2 - 18x + 5) \div (x + 2)$

$$\begin{array}{r} 5x^2 - 12x + 6 \\ x + 2 \overline{) 5x^3 - 2x^2 - 18x + 5} \\ \underline{-(5x^3 + 10x^2)} \\ -12x^2 - 18x \\ \underline{-(-12x^2 - 24x)} \\ 6x + 5 \\ \underline{-(6x + 12)} \\ -7 = \text{remainder} \end{array}$$

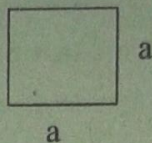
Thus, the quotient is $5x^2 - 12x + 6 - \frac{7}{x+2}$

MENSURATION OF PLANE FIGURES

1. Square

$$\text{Area} = (\text{edge})^2$$

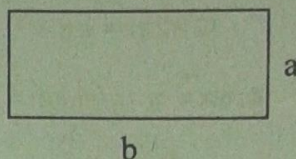
$$A = a^2$$



2. Rectangle

$$\text{Area} = \text{base} \cdot \text{altitude}$$

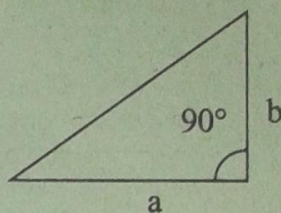
$$A = ba$$



3. Right triangle

$$\text{Area} = \frac{1}{2} \text{base} \cdot \text{altitude}$$

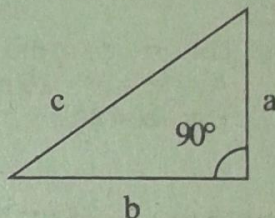
$$A = ba$$



Pythagorean Theorem

(Hypotenuse)² = sum of the squares of the two legs.

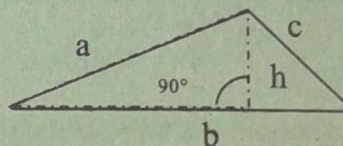
$$c^2 = a^2 + b^2$$



4. Oblique Triangle

$$\text{Area} = \frac{1}{2} \text{base} \cdot \text{altitude}$$

$$A = \frac{1}{2} bh$$

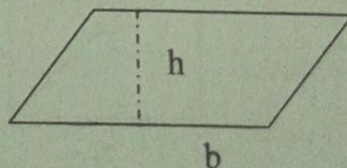


$$A = \sqrt{s(s-a)(s-b)(s-c)}, \text{ where } s = \frac{(a+b+c)}{2}$$

5. Parallelogram (opposite sides parallel)

$$\text{Area} = \text{base} \cdot \text{altitude}$$

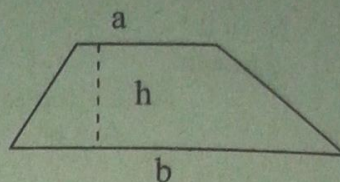
$$A = bh$$



6. Trapezoid (one pair of opposite sides parallel)

$$\text{Area} = \frac{1}{2} \text{ sum of base} \cdot \text{altitude}$$

$$A = \frac{1}{2}(a + b) h$$



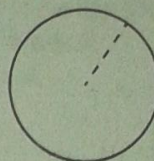
7. Circle

$$\text{Circumference} = 2\pi(\text{radius}) = \pi(\text{diameter})$$

$$C = 2\pi r = \pi d$$

$$\text{Area} = \pi (\text{radius})^2 = \frac{\pi}{4} (\text{diameter})^2$$

$$A = \pi R^2 = \frac{\pi}{4} D^2$$



EXERCISES

1. A square has a side of 5 inches. What is its area and perimeter?

Solution: Let x = side of the square

$$A = x^2 = 5^2 = 25 \text{ sq. inches}$$

$$P = 4x = 4(5) = 20 \text{ inches}$$

2. A square has an area of 400 sq;cm. What is its perimeter?

Solution: Let x = side of the square

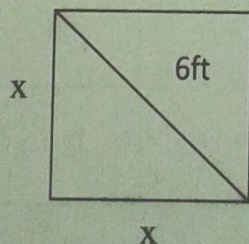
$$A = x^2 = 400$$

$$x = 20 \text{ cm}$$

$$P = 4x = 4(20) = 80 \text{ cm}$$

3. If the diagonal of a table with a square top is 6ft., what is the area of the table top?

Solution:



By Pythagorean theorem

$$X^2 + X^2 = 6^2$$

$$2X^2 = 36$$

$$X^2 = 18 \text{ sqf ft} = \text{Area}$$



4. A rectangular farm has dimensions of length 500m and width 400 m. Find its area in hectares.

Solution: $A = LW = (500)(400) = 20,000 \text{ sq. m.}$
 $A = 20,000 \text{ sq. m.} \times 1 \text{ hectares} = 2 \text{ hectares}$
 $10,000 \text{ sq. m.}$

5. One side of a rectangle is 20 inches. If the perimeter is 60 inches, what is the length of the other side?

Solution: Let x = unknown side
 $2x + 2(20) = 60$
 $2x = 20$
 $x = 10 \text{ inches}$

6. A square and an equilateral triangle have equal perimeters. What is the length of a side of a triangle if the area of the square is 9 square units?

Solution: Area of square = 9 sq. units
Side of the square = 3 units
Perimeter of the square = $4(3) = 12 \text{ units}$
Perimeter of the equilateral triangle = 12 units
Let x = side of the equilateral triangle
 $3x = 12$, therefore $x = 4 \text{ units.}$

7. The perimeter of an isosceles triangle is 42 inches. If one of the equal sides is 3 times the 3rd side, find the length of the 3rd side.

Solution: Let x = length of the 3rd side
 $3x + 3x + x = 42$
 $7x = 42$
 $x = 6 \text{ inches (3rd side)}$

8. The base of a container is a trapezoid. The lengths of the parallel opposite sides of the base are 6 inches and 12 inches with height 4 inches, Find the area of the base.

Solution:
 $A = \frac{1}{2} (6 + 12) 4 = 36 \text{ sq. inches}$

9. The diameter of a circle is 4ft, find its circumference and area.

Solution:
Diameter = 4ft Radius = 2ft
Circumference = $2\pi r = 2\pi (2) = 4\pi \text{ ft}$
Area = $\pi r^2 = \pi (2)^2 = 4\pi \text{ sq. ft.}$

10. The altitude of a right triangle is 4ft. Its hypotenuse is 5ft. Find the area and perimeter of the right triangle.

Solution: Let x = base of the right triangle
By Pythagorean theorem
 $x^2 + 4^2 = 5^2$ $x = 3\text{ft.}$
 $A = \frac{1}{2} (3) (4) = 6 \text{ sq. ft.}$
 $P = 3 + 4 + 5 = 12 \text{ ft.}$

INTRODUCTION TO PROBABILITY THEORY

Introduction

The study of probability is vital to us humans, for no day will pass without someone asking about probability or the chance that a certain event will occur. For example, before living home, one may ask about the chances that rain will occur. Another example is that of a close relative of a patient scheduled to undergo a risky operation asking about the survival chances of his kin. The study of probability therefore, involves the success or failure of certain events and as a key to its understanding, the number of ways by which events can happen should be studied.

Techniques of Counting

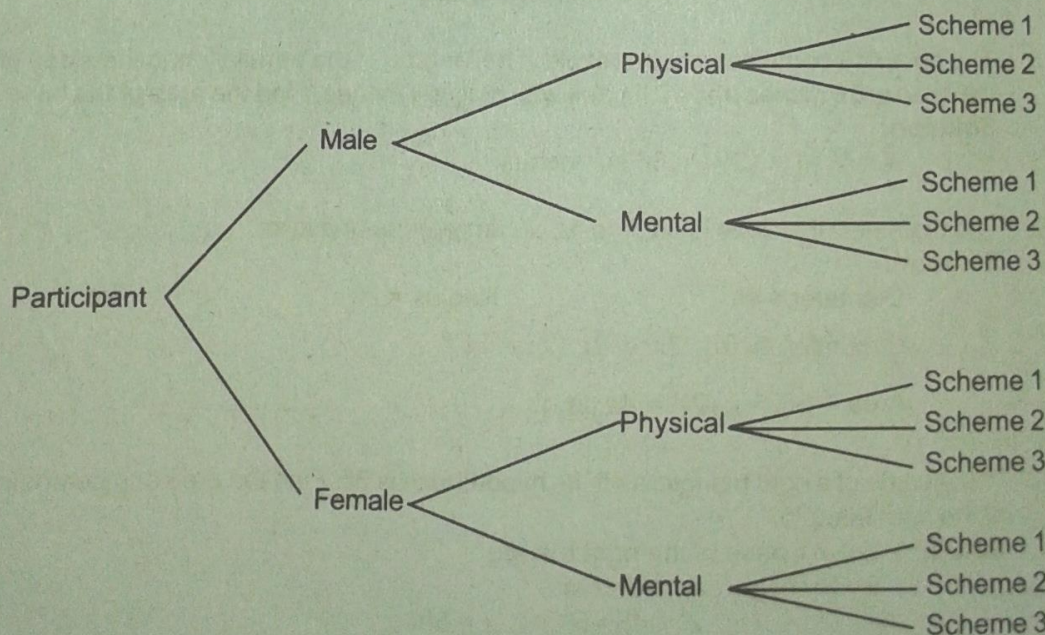
There are different techniques for counting the number of ways that events can happen, namely, the tree diagram, the fundamental principle of counting, permutation, and combination. These techniques could be used singly or in combination.

The Fundamental Principle of Counting

Given two events, E_1 and E_2 , if E_1 can happen in n_1 ways and E_2 can happen in n_2 ways, then the number of ways that both events can happen in the specified order n_1 is n_2 times.

Example: In a planned longitudinal study on the effect of 3 different weights reducing schemes, the following factors were also considered: gender of participant (male and female) and type of daily activity (physical, mental). In how many ways may the participants be classified? Do a tree diagram.

Example



The participants in the weight reducing scheme could be grouped into $2 \cdot 2 \cdot 3 = 12$ groups.

Math

Permutation

Permutation is an arrangement of objects wherein order is taken into account.

Factorial Notation

The expression $n!$ (read as n factorial) is equal to $n(n-1)(n-2)\dots(n-r+1)$ where $r \leq n$.

Example: $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$

$$5! = 120$$

By definition:

Proof: If $r = n$, then $n - r = 0$. Thus, the last term in the expansion will be $(0 + 1) = 1$.

Permutation of n objects taken all together

$${}_nP_n = n!$$

Example:

$$1. \quad {}_6P_6 = 6!$$

$$= 720$$

2. In how many different ways may four newly admitted patients be assigned to four available beds in the charity of a hospital?

$${}_4P_4 = 4! = 24$$

3. In how many different ways may seven victims of the cholera epidemic be assigned to a ward with seven beds if two of the victims who are sisters insist on being together?

$${}_6P_6 \cdot {}_2P_2 = 6! \cdot 2!$$

$$= 720(2)$$

$$= 1,440 \text{ ways}$$

*Note: The 2 sisters are considered as one group since they move together. However, they can interchange places by themselves.

Permutation of n objects r at a time

$${}_nP_r = \frac{n!}{(n-r)!}$$



Example:

$$\begin{aligned}
 1. \quad {}_6P_2 &= \frac{6!}{4!} \\
 &= \frac{6!}{(6-2)!} \\
 &= \frac{6 \cdot 5 \cdot 4!}{4!} \\
 &= 30
 \end{aligned}$$

2. In how many different ways may the chief of a medical mission to a disaster stricken area assign six out of nine doctors to head a team of health practitioners to six relief stations?

$$\begin{aligned}
 {}_9P_6 &= \frac{9!}{(9-6)!} \\
 &= \frac{362,880}{6} \\
 &= 60,480 \text{ ways}
 \end{aligned}$$

Permutation of n objects not all distinct

When one cannot distinguish between objects of the same class or characteristics, the appropriate formula to use is:

Where
$$\frac{N!}{n_1!n_2!\dots n_k!}$$

N = total number of objects

n = number of objects belonging to a particular class.

Example: In how many different ways can we arrange 10 experimental hamsters in 10 laboratory cages if 4 hamsters are white, 3 are brown and 3 are gray?

$$\frac{10!}{4!3!3!} = \frac{3,628,800}{864} = 4,200 \text{ ways}$$

Math

Circular Permutation

In circular permutation, one of the positions have to be fixed so that it will serve as the point of reference in counting the movement of the object in the circle. Thus, $(n-1)!$

$$\begin{aligned}\text{Example: 1. } (6-1)P_{(6-1)} &= (6-1)! \\ &= 5! \\ &= 120\end{aligned}$$

2. In a culture and sensitivity test, a microbiologist has to arrange circular strips of different antibiotics in a circular fashion on a large Petri dish with culture media and swab it with the microorganism being tested. In how many ways may the microbiologist implant 7 of these antibiotic strips in a circle?

$$\begin{aligned}(7-1)! \\ &= 6! \\ &= 720 \text{ ways}\end{aligned}$$

Combination

- Combination is an arrangement of objects where order or position is not important. The combination of n objects taken all together is 1.

$${}_nC_n = 1$$

$$\text{Example: } {}_5C_5 = 1$$

$${}_{100}C_{100} = 1$$

Combination of n object taken r at a time

$$\bullet \quad {}_nC_r = \frac{n!}{(n-r)!r!}$$

$$\begin{aligned}\text{Example: 1. } {}_7C_3 &= \frac{7!}{(7-3)!3!} \\ &= \frac{7!}{4!3!} \\ &= \frac{7 \cdot 6 \cdot 5 \cdot 4!}{4!3!} = 35\end{aligned}$$

3. On a survey on the effects of breastfeeding on infants it was decided by the researchers that 40% of nursing mother (those with infants being nursed) of every barangay in the study will be included in the sample. In a particular barangay, the researchers are to choose 4 out of 10 nursing mothers. In how many ways can this be done?

$$\begin{aligned} {}_{10}C_4 &= \frac{10!}{(10-4)!4!} \\ &= \frac{10!}{6!4!} \\ &= \frac{3,628,800}{(720)(24)} \\ &= 210 \text{ ways} \end{aligned}$$

Combination in a Series

$$\bullet {}_nC_1 + {}_nC_2 + \dots + {}_nC_n = 2^n - 1$$

Example:

In how many ways may a rural health officer assign one or more of 6 available barangay health workers to do community work?

$$2^6 - 1 = 63 \text{ ways}$$

Classical Definition of Probability

The classical (a priori) concept of probability is given by:

$$\Pr(E) = \frac{n}{N}$$

Where E is any given event

n is the number of successes of the event

N is the total number of possible outcomes of the event.

Example: In a family of 2 children, what is the probability of having exactly one boy?

To answer this problem, the total number of outcomes must be listed first. This **fundamental sample space** (S).

$$S = \{B_1B_2, B_1G_2, G_1B_2, G_1G_2\}$$

There are 4 possible outcomes (N) of which only two contain exactly 1 boy. Therefore if E_1 is the event of having exactly one boy then:

$$\Pr(E_1) = \frac{2}{4} = \frac{1}{2}$$

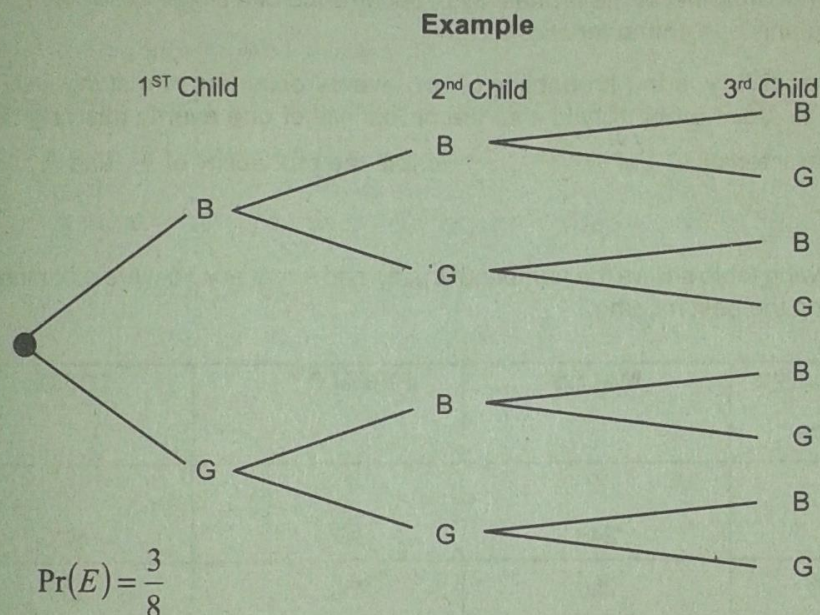
The probability of non-occurrence of an event E (called failure) is represented by the compliment of the event E . That is:

Example:

Let E be the event of having 2 boys in a family of 3 children.

$$S = \{BBB, BGB, GBB, BBG, BGG, GBG, GGB, GGG\}$$

There are eight possible outcomes. These outcomes could easily be obtained using the tree diagram.



The complement of E which is $\bar{E}$ means not getting exactly 2 boys in a family of 3 children and this given by:

$$\Pr(\bar{E}) = 1 - \Pr(E)$$

$$= 1 - \frac{3}{8}$$

$$= \frac{5}{8}$$

**Empirical Definition of Probability**

- **Empirical probability** (a posterior) refers to the relative frequency of the occurrence of an event when the number of trials is large. Its value is based on the actual occurrence of the event.

Example:

- Given an event which refers to the event of getting a blood type O in a mass blood typing activity of the Philippine National Red Cross, suppose 600 of the 1,000 persons who had their blood examined came out as:

$$\Pr(E) = \frac{600}{1,000} = 0.6$$

Marginal and Joint Probability

- **Marginal probability** is the probability of occurrence of a single event or an event satisfying only one characteristic.
- **Joint probability** is the probability of two events occurring simultaneously in a single trial. Joint probability is also the probability of one event satisfying two or more characteristics. Let $\Pr(E_1 \cap E_2)$ denote the probability of E_1 and E_2 .

Example:

- The following table shows the number of males and females who were examined for HIV during the past months.

SEX		MALES	FEMALES	TOTAL
RESULTS	HIV +	2	3	5
	HIV -	248	247	495
	TOTAL	250	250	500

The marginal probability for HIV + cases is:

$$\Pr(HIV +) = \frac{5}{500} = 0.01$$

The marginal probability for females is:

$$\Pr(F) = \frac{250}{500} = 0.5$$

The joint probability for females, who are HIV + is:

$$\Pr(E \cap HIV+) = \frac{3}{500} = 0.006$$

Mutually Exclusive Events

Two events E_1 and E_2 are mutually exclusive if the occurrence of E_1 precludes the occurrence of E_2 and vice versa. The joint probability of E_1 and E_2 in this case is zero. If and are capable of occurring simultaneously in a single trial then they are said to be non-mutually exclusive.

The Addition Law

The probability that an event or an event but not both will occur in a single trial given by:

$$1. \Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2)$$

For mutually exclusive events

$$2. \Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 \cap E_2)$$

For mutually exclusive events

Example:

1. In a certain hospital, 5 infants are fed with infant formula A, 4 infants are fed with infant formula B and 3 infants are fed with infant formula C. If one infant is selected at random, what is the probability he/she was fed with infant formula A or B?

$$\Pr(A \cup B) = P(A) + P(B)$$

$$= \frac{5}{12} + \frac{4}{12}$$

$$= \frac{9}{12}$$

$$= \frac{3}{4}$$

2. Of 10 medicines recommended for cough 6 were found to be effective, 4 were found to be expensive, and 2 were found to be effective and expensive. Find the probability that one medicine chosen at random is either effective or expensive.

$$\Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 \cap E_2)$$

$$= \frac{6}{10} + \frac{4}{10} - \frac{2}{10}$$

$$= \frac{8}{10} = \frac{4}{5}$$



3. The probability of 0, 1, 2, 3 children will be attended to by 3 school dentists at a particular time are 0.08, 0.22, 0.45 and 0.25 respectively. Find the probability that in the same time frame.

- a. Exactly 2 children are attended to
Let X = number of school children attended to

$$\Pr(X = 2) = 0.45$$

- b. No child is being attended to

$$\Pr(X = 0) = 0.08$$

- c. At least one child is being attended to

$$\Pr(x \geq 1) = 1 - \Pr(x = 0)$$

$$\begin{aligned}\text{or } \Pr(X \geq 1) &= P(X = 1) + \Pr(X = 2) + \Pr(X = 3) \\ &= 0.22 + 0.45 + 0.25\end{aligned}$$

Conditional Probability

The probability that an event, E_2 will occur that some event E_1 has already occurred is called **conditional probability**.

$$\Pr(E_1 | E_2) = \frac{\Pr(E_1 \cap E_2)}{\Pr(E_2)}$$

Example:

- The probability of randomly selecting a smoker (S) among 1,000 individuals is 0.50 and the probability of randomly selecting a person with lung cancer (C) among the same 1,000 individuals is 0.10. If the probability of selecting an individual who is both smoker and has lung cancer is 0.04, find the probability of selecting an individual who has lung cancer given that he is a smoker.

$$\begin{aligned}\Pr(C | S) &= \frac{\Pr(S \cap C)}{P(S)} \\ &= \frac{0.04}{0.50} \\ &= 0.08\end{aligned}$$

Dependent and Independent Events

Two events E_1 and E_2 are **independent** if the occurrence or non-occurrence of one does not affect the occurrence or non-occurrence of the other, otherwise, the event is **dependent**.



The probability that 2 events, E_1 and E_2 will occur one after the other in the stated order is given by the product of their respective probabilities.

$$1. \Pr(E_1 \cap E_2) = \Pr(E_1) \cdot \Pr(E_2)$$

... for independent events

$$2. \Pr(E_1 \cap E_2) = \Pr(E_1) \cdot \Pr(E_2 | E_1)$$

... for dependent events

Example 1:

By way of experience of a certain hospital, it was found out that 60% of their triple bypass operations for patients with heart ailments are successful. Find the probability that for the next three patients operated on:

- a. All three patients survive.

$$\begin{aligned} P(All) &= (0.60)(0.60)(0.60) \\ &= 0.216 \end{aligned}$$

- b. Exactly 2 survive.

$$\begin{aligned} P(X=2) &= (0.60)(0.60)(0.40) + (0.60)(0.40)(0.60) \\ &\quad + (0.40)(0.60)(0.60) \\ &= 0.432 \end{aligned}$$

Example 2:

In a sample of 10 school children, 4 were found to be malnourished. If the 2 children are selected one after another and sent to the school clinic for further examination, what is the probability that:

- a. Both are malnourished?

$$\begin{aligned} P(E_1 \cap E_2) &= P(E_1) \cdot P(E_2 | E_1) \\ &= \left(\frac{4}{10}\right)\left(\frac{3}{9}\right) \\ &= \frac{2}{15} \end{aligned}$$

- b. The first child is malnourished and the second child is not?

$$\begin{aligned} P(E_1 \cap \bar{E}_2) &= P(E_1) \cdot P(\bar{E}_2 | E_1) \\ &= \left(\frac{4}{10}\right)\left(\frac{6}{9}\right) \\ &= \frac{4}{15} \end{aligned}$$



1. The average of 5 students is 85. The average of the first 2 students is 88. What is the average of the other 3 students?

A. 81
B. 82

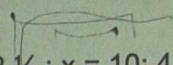
C. 83
D. 84

Answer: C

$$\begin{aligned} 85 \times 5 &= 425 \\ 425 - 176 &= 249 \end{aligned}$$

$$\begin{aligned} 88 \times 2 &= 176 \\ 249 \div 3 &= 83 \end{aligned}$$

2. Solve for x.


 $2 \frac{1}{2} : x = 10 : 4$

A. 2
B. 1

C. $\frac{1}{2}$
D. 3

Answer: B

Product of means = product of extremes

$$10x = \frac{5}{2}(4)$$

$$10x = 10$$

$$x = 1$$

3. 12.5% of 30 is what number?

A. 3.5
B. 3.75

C. 4.75
D. 4.5

Answer: B

$$.125(30) = 3.75$$

4. 0.12 is what percent of 24?

A. 0.5 %
B. 0.05%
C. 0.005%
D. 5%

Answer: A

$$0.12 \div 24 = .005 = 0.5\%$$

5. Last year, 2000 students enrolled in MET Review classes. This year, enrollment increased by 8%. How many students enrolled this year?

A. 2160
B. 2100
C. 2200
D. 2180

Answer: A

$$108\% \times 2000 = 1.08 \times 2000 = 2160 \text{ students}$$

6. A TV set that originally cost ₱15,000 is now sold at ₱12,000. What is the percentage decrease in the price of the TV set?

- $$\frac{\text{P}15,000 - \text{P}12,000}{\text{P}15,000} = 0.20 = 20\%$$

- $$\frac{5}{9} (180 \text{ degrees}) = 100 \text{ degrees}$$

- $$X = \frac{\text{P}180}{0.8} = \text{P}225$$

- $$75\% \text{ of } \text{P}1200 = 0.75 (\text{P}1200) = \text{P}900$$

- $$\frac{5}{8}(120) = 75 \text{ hrs.}$$



11. Three boys have marbles in the ratio of 19:5:3. If the boy with the least number has 9 marbles, how many does the boy with the greatest number have?

A. 48
B. 81
C. 60
D. 57

Answer: D

Let x = total number of marbles

$$\frac{3}{27}x = 9 \quad x = 81 \text{ marbles}$$

$$\frac{19}{27}(81) = 57 \text{ marbles}$$

12. Arnel borrowed ₱10,000 from the bank at the rate of 13% per year. How much will he owe the bank after a year?

A. ₱12,000
B. ₱11,300
C. ₱11,000
D. ₱11,400

Answer: B

$$\text{Interest} = ₱10,000 (0.13)(1) = ₱1300$$

$$\text{Amount due after a year} = ₱11,300$$

13. What is the total sales if for a 15% rate of commission, an agent gets ₱10,500?

A. ₱70,500
B. ₱80,500
C. ₱70,000
D. ₱80,000

Answer: C

$$\text{Total sales} = \frac{₱10,500}{15\%} = ₱70,000$$

14. A salesman gets 12% commission for the first ₱15,000 and 10% for the amount over ₱15,000 of his total sales. How much does he get for a total sales of ₱30,600?

A. ₱3400
B. ₱3360
C. ₱3340
D. ₱3460

Answer: B

$$\begin{aligned} \text{Total Commission} &= .12(₱15,000) + .10(₱15,600) \\ &= ₱1800 + ₱1560 = ₱3360 \end{aligned}$$

15. How much should a man deposit in a bank at 8% interest if he wants to gain ₱1,600 in a year?

A. ₱16,000
B. ₱15,000
C. ₱18,000
D. ₱20,000

Answer: D

$$\text{Principal} = \frac{\text{Interest}}{\text{rate (time)}} = \frac{₱1,600}{.08(1)} = ₱20,000$$

16. A car is able to travel 210 km in 3 hours. How far can it travel in 8 hours?

- A. 600 km
B. 560 km

- C. 500 km
D. 660 km

Answer: B

$$\text{Direct Proportion } \frac{x}{210} = \frac{8}{3}$$

$$x = 560 \text{ km}$$

17. Jean buys 3 apples for ₱50. How much does she pay for a dozen of these apples?

- A. ₱150
B. ₱220
C. ₱200
D. ₱180

Answer: C

$$1 \text{ dozen} = 12$$

$$12 \div 3 = 4$$

$$4 \times ₱50 = ₱200 \text{ (cost of 1 dozen of apples)}$$

18. To finish a certain job in 6 days, 8 workers are needed. If it is required to finish the same job 2 days earlier, how many workers have to work?

- A. 12
B. 10

- C. 6
D. 14

Answer: A

$$\text{Indirect Proportion } \frac{x}{8} = \frac{6}{4}$$

$$x = 12 \text{ workers}$$

19. An amount of ₱2,100 is to be divided among 3 children John, Jay and Joseph. Jay will receive double that of John and Joseph will receive double that of Jay. How much will Joseph receive?

- A. ₱6000
B. ₱8000
C. ₱1000
D. ₱1200

Answer: D. The ratio will be 1:2:4. Joseph will receive the greatest share. His share will be

$$\frac{4}{7} (\text{₱}2100) = \text{₱}1200$$

20. Manny can finish typing 5 pages in 8 minutes. How long will it take him to finish 12 pages?

- A. 19.2 minutes
B. 19 minutes
C. 18.2 minutes
D. 18 minutes

Answer: A

$$\text{Direct Proportion } \frac{x}{8} = \frac{12}{5}$$

$$x = 19.2 \text{ minutes}$$



21. A rectangular figure is 6 inches wide and 1 foot long. If the width is to be decreased by 2 inches, by how much should the length be increased, if the area remains the same?

A. 8 inches
B. 2 inches
C. 6 inches
D. 4 inches

Answer: C. Let x be the new length of the rectangle.

By indirect proportion, $\frac{x}{12} = \frac{6}{4}$

$x = 18$ inches. Therefore, the length should be increased by 6 inches

22. Which of these numbers does NOT belong to the group?

A. 83
B. 91
C. 71
D. 97

Answer: B. All are prime numbers except 91 which is composite. The divisors of 91 are 1, 91, 13 and 7.

23. Lito spent half of his money in SM and half of the remaining at Rustan Store. If he had 250 left, how much did he have at the start?

A. ₱ 1000
B. ₱ 1200
C. ₱ 1500
D. ₱ 1600

Answer: A

Let x = amount of money Lito have at the start

$$x - \left[\frac{1}{2}x + \frac{1}{4}x \right] = ₱ 250$$

$$\frac{1}{4}x = ₱ 250$$

$$x = ₱ 1000$$

24. Maria bought 120 handkerchiefs at 10 pesos each. Then she sold them at 3 handkerchiefs for ₱50. If she sold all the handkerchiefs, how much profit did she make?

A. ₱ 600
B. ₱ 800
C. ₱ 850
D. ₱ 650

Answer: B

$$120 \div 3 = 40$$

$$40 \times 50 = 2000 \text{ (gross profit)}$$

$$120 \times ₱10 = ₱1200 \text{ (cost)}$$

$$\text{Net profit} = ₱2000 - ₱1,200 = ₱800$$

Math

25. The price of one seat in a theatre is ₱40 and an average of 400 persons enter everyday. If the price was increased to ₱50, how many persons would yield the same amount to the theatre?

A. 300
B. 340

C. 350
D. 320

Answer: D

$$400 \times ₱40 = ₱16000$$
$$16000 \div 50 = 320 \text{ persons}$$

26. A man is 4 times as old as his son. Five years ago, the sum of their ages is 65. How old is the son?

A. 18
B. 15

C. 10
D. 12

Answer: B

Let x be the age of the son

$$(x-5) + (4x-5) = 65 \quad x = 15$$

27. A square has an area of 400 sq. cm. What is its perimeter?

A. 80 cm.
B. 60 cm.

C. 100 cm.
D. 120 cm.

Answer: A

Let x be the side of the square

$$A = x^2 = 400 \text{ sq. cm.} \quad x = 20 \text{ cm.}$$
$$P = 4x = 4(20 \text{ cm}) = 80 \text{ cm.}$$

28. The surface area of a cube is 150 sq. cm. What is the volume of the cube?

A. 64 cu. cm.
B. 300 cu. cm.

C. 81 cu. cm.
D. 125 cu. cm.

Answer: D

Let x be the side of the cube

$$SA = 6x^2 = 150 \quad x^2 = 25 \quad x = 5 \text{ cm.}$$
$$V = x^3 = 5^3 = 125 \text{ cu. cm.}$$

29. A rectangular sheet of paper is folded in half. Then it is folded again in half, and again in half. If the foldings in half is done 5 times, the edges are creased each time, how many small rectangles will you see when you unfold the original sheet of paper?

A. 16
B. 25

C. 32
D. 24

Answer: C

$$2^5 = 32$$



30. $\frac{1}{6}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{5}{6}, \text{---}$

- A. $\frac{3}{4}$
B. 1

- C. $\frac{7}{6}$
D. $\frac{1}{4}$

Answer: B. An arithmetic progression. Succeeding term is obtained by adding a fix number $\frac{1}{6}$.

That is $\frac{5}{6} + \frac{1}{6} = \frac{6}{6} = 1$.

31. If the diagonal of a table with square top is 6 ft., what is the area of the table top?

- A. 18 ft.
B. 18 sq. ft.

- C. 36 sq. ft.
D. 36 ft.

Answer: B

Let x be the side of the square top.

By Pythagorean theorem, $x^2 + x^2 = 6^2$.

$$2x^2 = 36$$

$$x^2 = 18 \text{ sq. ft.} = \text{Area of the table top}$$

32. If $m > 0$ and $n < 0$, which of the following is true?

A. $\frac{m}{n} = 0$

C. $\frac{1}{m} > \frac{1}{n}$

B. $mn > 0$

D. $\frac{1}{n} > \frac{1}{m}$

Answer: C

If $m > 0$, then $\frac{1}{m} > 0$

If $n < 0$, then $\frac{1}{n} < 0$

Therefore, $\frac{1}{m} > \frac{1}{n}$

33. Find the value of $(8x2)^{-3}$ when $x = 1$.

A. $\frac{1}{216}$

C. $\frac{1}{81}$

B. 216

D. 81

Answer: A

Substitute $x = 1$ to $(8x2)^{-3}$.

$$(8(1)-2)^{-3} = 6^{-3} = \frac{1}{6^3} = \frac{1}{216}$$

34. Simplify: $\left(\frac{x^2}{y}\right)^2 \left(\frac{y^2}{x}\right)^3$

A. $\frac{x}{y^4}$
B. $\frac{x^4}{y}$

C. xy^4

D. x^4y

Answer: C

$$\left(\frac{x^2}{y}\right)^2 \left(\frac{y^2}{x}\right)^3 = \frac{x^4}{y^2} \cdot \frac{y^6}{x^3} = xy^4$$

35. A, B and C are consecutive whole numbers. If $A > B > C$, what is the value of (A-B) (A-C) (B-C)?

- A. 1
B. -1
C. 2
D. -2

Answer: C

$$A - B = 1 \quad A - C = 2 \quad B - C = 1$$

Therefore $(A-B) (A-C) (B-C) = (1)(2)(1) = 2$.

36. If x is an odd integer and y is an even integer, which of the following is an odd integer?

- A. $2x-y$
B. $x^2 + y-1$
C. $x^2 + 3y$
D. $x-1$

Answer: C

$$x^2 + 3y = (\text{odd})^2 + 3(\text{even}) = \text{odd} + \text{even} = \text{odd}$$

37. Simplify: $12\frac{1}{8} - 3\frac{5}{6} - 7\frac{2}{3}$

- A. $\frac{1}{8}$
B. $1\frac{5}{8}$
C. $\frac{5}{8}$
D. $\frac{3}{8}$

Answer: C

$$\begin{aligned} 12\frac{1}{8} - 3\frac{5}{6} - 7\frac{2}{3} &= (12 - 3 - 7) + \left(\frac{1}{8} - \frac{5}{6} - \frac{2}{3}\right) \\ &= 2 + \frac{3 - 20 - 16}{24} = 2 + \left(\frac{-33}{24}\right) \\ &= 1\frac{24}{24} - 1\frac{9}{24} = \frac{15}{24} = \frac{5}{8} \end{aligned}$$

38. Simplify: $(3x+5) - [3x - (-x + 1) - 3]$

- A. $-x - 9$
- B. $-9 + x$
- C. $9 + x$
- D. $9 - x$

Answer: D

$$\begin{aligned} & (3x + 5) - [3x - (-x + 1) - 3] \\ & = 3x + 5 - [3x + x - 1 - 3] \\ & = 3x + 5 - (4x - 4) = 3x + 5 - 4x + 4 = -x + 9 = 9 - x \end{aligned}$$

39. A dinner menu offers choices of 5 appetizers, 8 main dishes and 2 desserts. How many ways can you choose one of each course?

- A. 15
- B. $5!8!2!$
- C. $15!$
- D. 80

Answer: D

Techniques of counting. $5(8)(2) = 80$ ways

40. A ball is drawn at random from a box containing 3 red balls, 4 white balls and 5 blue balls. Find the probability that the ball is NOT red.

- A. $\frac{1}{4}$
- B. $\frac{3}{4}$
- C. $\frac{2}{3}$
- D. $\frac{1}{3}$

Answer: B

$$P = \frac{\text{no. of balls not red}}{\text{total no. of balls}} = \frac{4 + 5}{3 + 4 + 5} = \frac{9}{12} = \frac{3}{4}$$

41. When a pair of fair dice is thrown, what is the probability of getting a sum of 9?

- A. $\frac{2}{3}$
- B. $\frac{1}{9}$
- C. $\frac{2}{9}$
- D. $\frac{1}{3}$

Answer: B

$$P = \frac{\text{no. of possible outcomes with sum 9}}{\text{total no. of possible outcomes}}$$

$$P = \frac{\{(3,6), (4,5), (5,4), (6,3)\}}{\{(1,1), (1,2), \dots, (2,1) \dots (2,6) \dots (6,6)\}}$$

$$P = \frac{4}{36} = \frac{1}{9}$$

Math

42. Simplify $\frac{4^{20}}{2^{40}}$

- A. 1
B. 4

- C. 2^{20}
D. $\left(\frac{1}{2}\right)^{20}$

Answer: A

$$\frac{4^{20}}{2^{40}} = \frac{(2^2)^{20}}{2^{40}} = \frac{2^{40}}{2^{40}} = 1$$

$$4 = 2^2$$

43. A big garden has a triangular shape. It has a height of 12 meters and a base of 9 meters. What is its area?

- A. 54 m^2
B. 42 m^2

- C. 122 m^2
D. 108 m^2

Answer: A

$$A = \frac{1}{2}bh = \frac{1}{2}(9)(12) = 54 \text{ m}^2$$

44. How many ounces of pure acid must be added to 20 ounces of a solution that is 5% acid to make a mixture that is 24% acid?

- A. 5
B. 3

- C. 10
D. 2

Answer: A

Let x be the no. of ounces of pure acid

$$100\%(x) + 5\%(20) = 24\%(x+20)$$

$$x + .05(20) = .24(x + 20)$$

$$x + 1 = .24x + 4.8$$

$$x - .24x = 4.8$$

$$.76x = 3.8$$

$$x = 5 \text{ ounces pure acid}$$

45. In how many ways can a coach assign the 5 starting positions in basketball to 10 equally qualified men?

- A. 252
B. 30, 240

- C. 5
D. 10

Answer: B. Permutation of 10 taken 5 at a time. $= {}_{10}P_5 = 30, 240$

46. In how many ways can 8 guests be seated in a round table with 8 chairs?

- A. $8!$
B. $7!$
C. $5!$
D. $6!$

Answer: B. Circular permutation. $(n-1)! = (8-1)! = 7!$



47. In how many ways may a rural health officer assign one or more of 6 available barangay health workers to do community work?

- A. $6!$
- B. 6
- C. $5!$
- D. 63

Answer: D. Combination in series. $2^n - 1 = 2^6 - 1 = 63$

48. A committee of 5 persons must be selected from a group of 5 men and 8 women. In how many ways may the selection be made with no restrictions?

- A. 154440
- B. 1287
- C. ${}_{13}P_5$
- D. $5!13!$

Answer: B. Combination of 13 taken 5 at a time = ${}_{13}C_5 = 1287$

49. The value of k for which the number 128k0 is divisible by 4.

- A. 1
- B. 3
- C. 5
- D. 8

Answer: D. A number is divisible by 4 if the number formed by the last 2 digits is divisible by 4. The value of k is 8 because 80 is divisible by 4. Therefore, 12880 is divisible by 4.

50. One side of a rectangle is x inches. If the perimeter is P inches, what is the length (in inches) of the other side?

- A. $\frac{2x+p}{2}$
- B. $\frac{p-2x}{2}$
- C. $\frac{p+x}{2}$
- D. $\frac{2x-p}{2}$

Answer: B

Let y be the other side of the rectangle

$$2x + 2y = P$$

$$2y = P - 2x$$

$$\frac{p-2x}{2} = y$$

51. 36.6 % in terms of fraction.

- A. $\frac{183}{1000}$
- B. $\frac{366}{500}$
- C. $\frac{366}{100}$
- D. $\frac{183}{500}$

Answer: D

$$36.6\% = .366 = \frac{366}{1000} = \frac{183}{500}$$

Math

52. $2\frac{1}{6}$ in terms of percent

- A. 216.6%
B. $216\frac{1}{3}\%$
C. $216\frac{2}{3}\%$
D. 216%

Answer: C

$$2\frac{1}{6} = 2.1666666... = 216.666666...% = 216\frac{2}{3}\%$$

53. ₱ 12,500 is what percent of ₱ 50,000?

- A. 10%
B. 15%
C. 25%
D. 20%

Answer: C

$$\text{₱}12,500 = R (\text{₱} 50,000) \quad R = .25 = 25\%$$

54. ₱ 15800 decreased by 12.5% is how much?

- A. ₱ 13,800
B. ₱ 13,525
C. ₱ 13,285
D. ₱ 13,825

Answer D

$$\text{₱}15,800 - 0.125 (\text{₱}15,800) = \text{₱} 13825$$

55. $\frac{3}{4}$ of $\frac{1}{3}$ of 30 is what % of 60?

- A. 15.625%
B. 25.89%
C. 26.89%
D. 25.79%

Answer: A

$$\frac{3}{4} \left(\frac{1}{3} \right) 30 = R \left(\frac{4}{5} \right) (60) \quad R = 0.15625 = 15.625\%$$

56. What is $16\frac{2}{3}\%$ of 42?

- A. 7
B. 6
C. 5
D. 7.5

Answer: A

$$16\frac{2}{3}\% (42) = 7$$



57. John jogs every morning. Yesterday, he covered 2 km. Today he covers 2.5 km. What is the percent of increase in the distance he traveled?

- A. 50%
- B. 25%

- C. 10%
- D. 20%

Answer: B

$$\frac{2.5 - 2}{2} = 0.25 = 25\%$$

58. A 180 – degree angle is divided into 3 in the ratio 1:2:3. What is the measure of the biggest angle?

- A. 45°
- B. 90°

- C. 60°
- D. 30°

Answer: B

$$\frac{3}{6} (180^\circ) = 90^\circ$$

59. What is the area of a rectangle of width 10 inches and perimeter of 120 inches?

- A. 200 sq. in.
- B. 300 sq. in.

- C. 400 sq. in.
- D. 500 sq. in.

Answer: D

$$\begin{aligned} P &= 2W + 2L \\ &= 2(10) + 2L \\ L &= 50 \text{ inches} \\ A &= WL = 10(50) = 500 \text{ sq. in.} \end{aligned}$$

60. Jay invested ₱ 4,500 in a buy and sell. How much will his money be in 3 years if it earns 12.5% simple interest rate in a year?

- A. ₱6,187.50
- B. ₱6,188
- C. ₱6,187
- D. ₱6,189.50

Answer A: $F = P(1+rt) = ₱4,500(1+0.125(3)) = ₱6,187.50$

61. A store allows ₱ 15 discount on all items. How much does a customer pay for an item which is marked ₱ 800?

- A. ₱700
- B. ₱790
- C. ₱680
- D. ₱785

Answer: D

$$₱800 - ₱15 = ₱785$$

Math

62. What is the percentage commission if a salesman gets ₱ 800 from a total sale of ₱15,000?

A. 5%

C. $5\frac{1}{3}\%$

B. 5.5%

D. 5.3%

Answer: C

$$\frac{800}{15000} = 5\frac{1}{3}\%$$

63. The side of a square of area $x^2 - 4x + 4$

A. $x - 4$

C. $x - 2$

B. $x + 4$

D. $x + 2$

Answer: C

$$\begin{aligned}\text{Area} &= x^2 - 4x + 4 \\ &= (x-2)^2 = (\text{side})^2 \\ &= x - 2\end{aligned}$$

64. Five liters of gasoline can travel a distance of 70 km. How far can 8 liters reach?

A. 80 km.

B. 50 km.

C. 100 km.

D. 112 km.

Answer: D

$$\frac{5}{8} = \frac{70}{x} \quad x = 112\text{km}$$

65. At 75 kph. Jigs can reach home within 50 minutes. At what rate should he drive his car so that he can reach home 10 minutes earlier?

A. 80 kph

C. 60 kph

B. 93.75 kph

D. 90 kph

Answer: B

$$\frac{75}{x} = \frac{40}{50} \quad x = 93.75 \text{ kph}$$

66. The area of a circle with circumference 4π

A. 4π

C. 6π

B. 2π

D. 3π

Answer: A

$$\begin{aligned}\text{circumference} &= 2\pi r = 4\pi & r &= 2 \\ A &= \pi r^2 = \pi(2)^2 = 4\pi\end{aligned}$$



67. Find X: $\frac{2\frac{1}{3}}{X} = \frac{1\frac{1}{6}}{6}$

- A. 4
B. 10

- C. 6
D. 12

Answer: D

$$\frac{7}{3x} = \frac{7}{36} \quad x = 12$$

68. An angle which measures between 0° and 90°

- A. Acute
B. Obtuse

- C. Right
D. Supplementary

Answer: A

69. Two angles whose sum of degrees is equal to 180°

- A. Complimentary
B. Supplementary
C. Obtuse
D. Vertical

Answer: B

70. Two complementary angles are in the ratio 1:4. What is the smaller angle?

- A. 40°
B. 18°

- C. 72°
D. 36°

Answer: B

$$\frac{x}{90-x} = \frac{1}{4} \quad x = 18^\circ$$

71. An angle which measures greater than 90° but less than 180° .

- A. Acute
B. Supplementary
C. Right
D. Obtuse

Answer: D

72. In an isosceles right triangle, the hypotenuse is 12cm long. What is the length of one of its congruent legs?

- A. $5\sqrt{2}$ cm
B. 5 cm

- C. $6\sqrt{2}$ cm
D. 6 cm

Answer: C

$$x^2 + x^2 = (12)^2 \quad x = 6\sqrt{2} \text{ cm}$$

73. A rectangular farm has dimensions 500 meters by 40 meters. Find its area in hectares.

- A. 2
- B. 3
- C. 2.5
- D. 3.5

Answer: A

$$A = (500)(40)m^2 = 20,000m^2 = 2 \text{ hectares}$$

74. What is the area of a triangle of sides 4, 7 and 9 units?

- A. 6 sq. units
- B. 5 sq. units
- C. $\sqrt{6}$ sq. units
- D. $\sqrt{5}$ sq. units

Answer: D

Using Heron's Formula

$$A = \sqrt{10(10-4)(10-7)(10-4)} = 6\sqrt{5} \text{ sq. units}$$

75. The circumference of a circle of area 4π .

- A. 4π
- B. π
- C. 2π
- D. 3π

Answer: A

$$\text{Area} = \pi r^2 = 4\pi$$

$$r = 2$$

$$\text{Circumference} = 2\pi r$$

$$= 2\pi(2) = 4\pi$$

76. The area of a circle of diameter 10 units.

- A. 25 sq. units
- B. 25π sq. units
- C. 5 sq. units
- D. 5π sq. units

Answer: B

$$A = \pi r^2 = \pi(5)^2 = 25\pi \text{ sq. units}$$

77. The area of a triangular garden of base 20 ft. and altitude 10 ft.

- A. 50 sq. ft.
- B. 10 sq. ft.
- C. 80 sq. ft.
- D. 100 sq. ft.

Answer:

$$A = \frac{1}{2}(20)(10) = 100 \text{ sq. ft.}$$

78. One of the legs of a right triangle is 5 cm long and its hypotenuse is 15 cm. How long is the other leg?

A. 10 cm

C. 8 cm

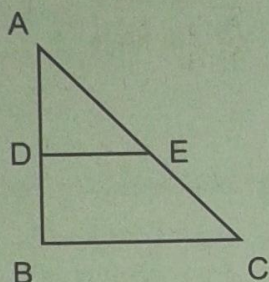
B. $10\sqrt{2}$ cm

D. $8\sqrt{2}$ cm

Answer: B

$$x = \sqrt{15^2 - 5^2} = \sqrt{200} = 10\sqrt{2} \text{ cm}$$

79. In the figure, $\triangle ADE \sim \triangle ABC$. If $AD = 6$, $DE = 9$ and $AB = 10$, what is BC ?



$$\frac{6}{9} = \frac{10}{x}$$

$$6 \cdot 9 = 10 \cdot x$$

$$x = 15$$

A. 8

C. 15

B. 10

D. 12

Answer: C

$$\frac{AD}{DE} = \frac{AB}{BC}$$

$$BC = 15$$

80. What type of a regular polygon is one whose interior angle measures 108 degrees?

A. Heptagon

B. Pentagon

C. Octagon

D. Hexagon

Answer: B

$$\frac{(n-2)180^\circ}{n} = 108^\circ$$

$$n = 5 \text{ pentagon}$$

81. What is the sum of the measure of the angles of a convex octagon?

A. 900°

B. 1200°

C. 1800°

D. 1080°

Answer D

$$(n-2)180^\circ = (8-2)180^\circ = 1080^\circ$$

Math

82. What is the average of $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{6}$?

- A. $\frac{1}{4}$
B. $\frac{1}{2}$

- C. $\frac{3}{4}$
D. $\frac{4}{3}$

Answer: A

$$\left(\frac{1}{3} + \frac{1}{4} + \frac{1}{6}\right) \div 3 = \frac{1}{4}$$

83. A meter stick was cut into 2 pieces at the 36cm – mark. What is the ratio of the smaller piece to the larger piece?

- A. 16:25
B. 13:50

- C. 9:25
D. 9:16

Answer: D

$$36:64 = 9:16$$

84. $2+3P$, $4+2P$, $6+P$, _____

- A. $4 + P$
B. 8
C. $4 - P$
D. 6

Answer: B

Add 2-p

85. $\frac{1}{2}$, $\frac{2}{3}$, $\frac{5}{6}$ _____

- A. 2
B. $\frac{1}{6}$

- C. 1
D. $\frac{5}{6}$

Answer: C

$$\text{Add } \frac{1}{6}$$

86. Find the quotient: $\left(\frac{3}{4} - \frac{2}{3}\right) \div \left(\frac{5}{6} - \frac{3}{5}\right)$

- A. $\frac{1}{14}$
B. $\frac{4}{14}$

- C. $\frac{3}{14}$
D. $\frac{5}{14}$

Answer: D

$$\left(\frac{3}{4} - \frac{2}{3}\right) \div \left(\frac{5}{6} - \frac{3}{5}\right) = \left(\frac{9-8}{12}\right) \div \left(\frac{25-18}{30}\right) = \frac{1}{12} \cdot \frac{30}{7} = \frac{5}{14}$$



87. A, B and C are all working in a certain company, they are all on leave. A will file a leave every 4 days, B every 6 days and C every 8 days. When will they be all on leave?

A. After 24 days
B. After 16 days

C. After 20 days
D. After 18 days

Answer: A LCM of 4, 6, and 8 is 24

88. Juan drives $5\frac{3}{4}$ km. to work and drives home the same day, for five working days a week, how many kilometers does he drive each week?

A. 57 km
B. 57.5 km
C. 58 km
D. 57 km

Answer: B

57.5 km

89. The average of 4 numbers is 28. If 3 of the 4 numbers are 18, 28, and 36, what is the 4th number?

A. 30
B. 28

C. 26
D. 24

Answer: A

$$3(28) = 112 \quad 18+28+36 = 82 \quad 112-82 = 30$$

90. The polynomial $x^2 + 4x - 32$ is the area of a rectangular floor. What is the length of one of its sides?

A. $x-2$
B. $x+4$

C. $x+8$
D. $x+2$

Answer: C

$$x^2+4x-32=(x-4)(x+8)$$

$x+8$ is one of the sides

91. $a^0 = ?$ $a \neq 0$

A. Undefined
B. Indeterminate

C. 0
D. 1

Answer: D

92. LCM of $4x^2y$ and $30xy^3$

A. $30x^2y^3$

B. $30x^3y^2$

C. $60x^2y^3$

D. $60x^3y^2$

Answer: C

$$\text{LCM of } 4x^2y \text{ and } 30xy^3 = 60x^2y^3$$

Math

93. Solve for x: $4(x-1) - 2(x+1) = 3x - 2$

- A. -4
B. 4

- C. 2
D. -2

Answer: A

$$4(x-1) - 2(x+1) = 3x - 2$$

$$4x - 2x - 4 - 2 = 3x - 2$$

$$2x - 3x = -2 + 4 + 2$$

$$-x = 4$$

$$x = -4$$

94. Factored form of $x^5 - x^3 + x^2 - 1$

A. $(x+1)(x-1)(x^2-x+1)$

C. $(x+1)(x-1)^2(x^2-x+1)$

B. $(x+1)^2(x-1)(x^2+x+1)$

D. $(x+1)^2(x-1)(x^2-x+1)$

Answer: D

$$x^5 - x^3 + x^2 - 1 = x^3(x^2 - 1) + 1(x^2 - 1) = (x^3 + 1)(x^2 - 1) = (x + 1)^2(x - 1)(x^2 - x + 1)$$

95. Simplify $4\sqrt{8} - 9\sqrt{18} - 3\sqrt{50}$

A. $34\sqrt{2}$

C. $18\sqrt{2}$

B. $-34\sqrt{2}$

D. $-18\sqrt{2}$

Answer: B

$$4\sqrt{8} - 9\sqrt{18} - 3\sqrt{50} = 4\sqrt{4(2)} - 9\sqrt{9(2)} - 3\sqrt{25(2)} = 8\sqrt{2} - 27\sqrt{2} - 15\sqrt{2} = -34\sqrt{2}$$

96. The sum of 2 consecutive numbers is 75. If x is the smaller number, what is the value of x.

- A. 35
B. 40

- C. 38
D. 37

Answer: D

$$x + (x+1) = 75$$

$$x = 37$$

97. The graph of linear equation is _____.

- A. Curve
B. Vertical

- C. Parabola
D. Line

Answer: D. Linear equation is a 1st degree equation which is a line.

98. The GCD of 18, 24 and 30

- A. 6
B. 8

- C. 4
D. 2

Answer: A

$$\text{GCD of } 18, 24, 30 = 6$$

99. The coin is tossed 3 times. What is the probability that not all 3 tosses are the same?

A. $\frac{1}{4}$

C. $\frac{5}{8}$

B. $\frac{3}{8}$

D. $\frac{3}{4}$

Answer: D

$n = 8$

$h = 6$

$p = \frac{6}{8} = \frac{3}{4}$

100. The tens digit of a certain two – digit number is 4 more than the unit's digit, the sum of the squares of the two digits is 26. Find the numbers.

A. 51

C. 45

B. 15

D. 54

Answer: A

Let x = units digit

y = tens digit

$y = x + 4$

$x^2 + y^2 = 26$

solving simultaneously 2 equation in 2 unknowns

$x = 1$ and $y = 5$.

The number is 51.